

PHASE 1: Virtual Machine Setup

Step 1: Install VirtualBox on Your Mac

- Download VirtualBox from <https://www.virtualbox.org>
- Install the Extension Pack for USB/network support

PHASE 2: Install ParrotOS in VirtualBox

Step 2: Set Up ParrotOS VM

- Download Parrot Security OS ISO from parrotsec.org
- Create a new VM in VirtualBox (Linux, Debian 64-bit)
- Allocate 4+ GB RAM, 50+ GB disk, boot ISO and install ParrotOS

PHASE 3: Install SOC Components in ParrotOS

Step 3: Install Wazuh + ELK Stack

- Run: `curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh`
- Then: `sudo bash wazuh-install.sh -i`
- Wazuh, Elasticsearch, Logstash, Kibana installed

PHASE 3: Configure Wazuh

Step 4: Access Wazuh Dashboard

- In ParrotOS: Go to <https://localhost> in browser
- Login with your credentials
- Ensure Kibana and services are running

PHASE 4: Log Collection from macOS

Step 5: Install Wazuh Agent on macOS

- Download pkg from Wazuh site
- `sudo installer -pkg wazuh-agent-4.x.x.pkg -target /`
- Point it to ParrotOS IP in `/Library/OSsec/etc/ossec.conf`
- Start agent

PHASE 5: Monitoring & Forensics Tools

Step 6: Install Zeek

- `sudo apt install zeek`
- `sudo zeek -i eth0`

Step 7: Install Wireshark

- sudo apt install wireshark
- sudo usermod -aG wireshark \$USER

PHASE 5: Velociraptor

Step 8: Install Velociraptor

- Download from GitHub
- chmod +x and move to /usr/local/bin
- Generate config: velociraptor config generate > server.config.yaml
- Run frontend

PHASE 6: Connecting Everything

Step 9: Connect Tools

- Ensure all services (Wazuh, Kibana, Elastic, Logstash) are running
- Check Kibana Discover tab for logs from ParrotOS and macOS

PHASE 7: Detection, Response, Reporting

Step 10: Create Detection Rules in local_rules.xml

Step 11: Simulate Attacks (SSH brute-force, PowerShell)

Step 12: Investigate with Velociraptor & Wireshark, write incident report

Full System Flow

macOS + ParrotOS Agents -> Wazuh Manager -> Logstash -> Elasticsearch -> Kibana
+ Zeek + Velociraptor -> You Analyze, Contain, Report